

GPS Engine board SR-87

SiRF StarIII

SR-87

V1.0.1



ProGin Technology Inc.

12F-1, No. 5, Lane 7, De-an St., Lingya District, Kaohsiung 802, Taiwan

Tel: +886-7-7278885

Fax: +886-7-7214117

E-mail: info@progin.com.tw <http://www.progin.com.tw>



***Description**

The ProGin SR-87 series GPS module incorporates high sensitivity, low power consumption SiRF StarIII chipset solution in a compact design. The module will track up to 20 satellites at the time while offering fast time-to-first-fix, 1Hz navigation update. The unit is very suitable for broad applications such as Handheld, PDA, PPC or other battery operated navigation system.

***Features**

- High sensitivity SiRF StarIII chipset.
- High performance receiver tracks up to 20 satellites.
- TTL output for GPS command interface.
- Low power consumption.
- Average Cold Start time under 42 seconds.
- On-chip 1Mb SRAM.
- Reacquisition time 0.1 second.
- Support accurate 1PPS output signal aligned with GPS timing.
- Support Standard NMEA-0183 and SiRF Binary protocol.
- Multi-path mitigation hardware.
- A Built-in Lithium battery enables fast positioning.
- Compact size (25.4×25.4×7 mm³) for easy integration into hand-held devices.

***Ordering information:**

SR-87V, Vertical MMCX input

SR-87H, Horizontal MMCX input

***Technical Specifications**

Electrical Characteristics

General

Frequency	L1, 1575.42 MHz
C/A code	1.023 MHz chip rate
Channels	20 channels all in view tracking



SR-87 GPS Engine Board Spec.

Sensitivity

Tracking -159 dBm typical

Accuracy (Open Sky)

Position < 10 meters, 2D RMS
< 7 meters 2D RMS, WAAS corrected
1-5 meters, DGPS corrected
Time 1 microsecond synchronized to GPS time

Datum

Default WGS-84

Acquisition Rate

Hot start 1 sec, average
Warm start 38 sec, average
Cold start 42 sec, average
Reacquisition 0.1 sec, average

Dynamic Conditions

Altitude < 18,000 meters (60,000 feet)
Velocity < 515 meters/sec (1000 knots)
Acceleration < 4 G
Jerk 20 meters/sec max

Power

Main power input 3.0 ~ 5.5 VDC input.
Supply Current < 80 mA
Backup Power 3V rechargeable Lithium battery,
up to 500 hours discharge

RF Interface

Antenna connector type MMCX



SR-87 GPS Engine Board Spec.

Serial Port

Electrical interface	Two full duplex serial communication, via RS232, TTL interface
Protocol message	NMEA-0183, SiRF Binary
Default NMEA	GGA, GSA, GSV, RMC, (GLL, VTG, and ZDA optional) 4800 baud rate, other rate optional 8 bits data, 1 stop bit, no parity.

Time

1 PPS Pulse, pulse duration 100 msec.
Time reference at the pulse positive edge.
Synchronized to GPS time, ± 1 microsecond.

Weight

< 8g

Recommended External Antenna Specification

Gain	20 dB min (cable loss included)
Noise figure	1.5 dB typical
Current	10 mA typical
Operating Voltage	Confirmed to spec 3.3 ~ 5.5 V
Survival	3.0 ~ 3.3 V

Environmental Characteristics

Operating temperature range	-40 °C to +85 °C
Storage temperature range	-45 °C to +100 °C

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